# NES AI FINANCE & INSURANCE INTELLIGENCE PLATFORM

Test Strategy & Acceptance Checklist

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**Build approach:** Cline-assisted development • Local-first Windows application

**Scope:** Personal research, paper trading and insurance document assistance

Quality gates, regression tests and data integrity

## 1. Test objectives

- Prevent misleading financial displays and workflow state errors.
- Ensure calculations are reproducible and assumptions visible.
- Verify local-first and simulation-only boundaries.
- Catch navigation, responsive, accessibility and animation regressions.

## 2. Test levels

| Level | Coverage |
|---|---|
| Unit | Financial calculations, validators, formatters, rule logic and state transitions. |
| Integration | API schemas, provider adapters, document parsing, database transactions. |
| UI/component | Forms, tables, modal behaviour, loading/error states and accessible controls. |
| End-to-end | Navigation, screener, backtest wizard, paper order simulation, policy review and comparison. |
| Security/privacy | Secret scanning, log redaction, file validation, local data access and backup handling. |
| Performance | Initial render, large tables, chart interaction, reduced motion and 3D fallback. |

| Level | Coverage |
| --- | --- |
| Accessibility | Keyboard-only use, focus order, labels, contrast and reduced motion. |

# 3. Acceptance checklist

- All six primary navigation destinations work.
- No button looks active if it has no meaningful action.
- All sample data is labelled illustrative/demo.
- Dark trading and light insurance themes remain readable.
- All 12 animations replay and reduced motion is respected.
- Core UI works without WebGL.
- Paper-trading action never calls a broker or creates a real order.
- Policy extraction requires review before verified status.
- Backtest report includes period, data source, fees/slippage assumptions and limitations.
- Provider failure, stale data and missing data are clearly distinguished.
- Backup restore is tested before storing real policy documents.

# 4. Financial calculation tests

- Unit tests for indicator calculations and edge cases.
- Tests for missing bars, duplicate timestamps, split/corporate-action assumptions where supported.
- Backtest fees, slippage, trade timing and benchmark methodology tested against known examples.
- No look-ahead bias or accidental future data leakage.
- Risk metrics validated with small manually checkable datasets.

# 5. Document and AI tests

- Test PDFs with selectable text, scanned pages, malformed files and mixed layouts.
- Test incorrect/missing dates, policy numbers, premiums and exclusions.
- Ensure extracted fields preserve source location or provenance when available.
- Test Copilot questions with missing evidence and conflicting documents.
- Ensure answers admit uncertainty and do not invent insurer terms or market facts.

# 6. Release gates

**Do not connect real market data, use real personal documents or rely on AI outputs until the relevant domain tests and privacy controls pass.**